

Arrow Wisconsin Technology Expo – Technical Seminar Descriptions*

Tuesday, October 13th, 2026

8:30am to 2:20pm

(Revision 2026-09-30)

8:30am to 9:20am – Session #1

NXP Semiconductor Presents:

Title: NXP Advanced Analog offering: interface IC's, AFE, including automotive and industrial application IC's

Abstract: "This seminar will provide an overview of NXP's Advanced Analog product portfolio, showcasing a broad range of Interface ICs, Analog Front Ends (AFEs), and automotive and industrial application-specific solutions designed to address the growing demands of connectivity, sensing, control, electrification, and system intelligence. The presentation covered key product areas including temperature sensors; GPIO expanders, bridges, I²C interfaces, voltage level translators, and SD card solutions; high-speed switches and redrivers; USB interface devices; power management ICs; real-time clocks (RTCs); LED and LCD controllers; and AC-DC power solutions. Attendees gained insight into how these products enable robust, efficient, and reliable system architectures across a diverse set of embedded applications.

The seminar will also highlight NXP's leadership in automotive and industrial analog technologies through its portfolio of T1/T1S Ethernet, CAN, CAN-FD, and LIN networking solutions; automotive switches; motor drivers; battery management systems and EV charging technologies; extreme switches serving as fuse and relay replacements; SiC and IGBT gate drivers; high-performance AFEs; and automotive lighting solutions. Through application examples, the presentation will illustrate how NXP's advanced analog offerings help engineers improve system performance, reduce design complexity, enhance safety and reliability, and accelerate the development of next-generation automotive, industrial automation, energy management, and electrification platforms."

onsemi Presents:

Title: Advancing Power Conversion with onsemi Lateral GaN Technology ranging from 40V-650V

Abstract: onsemi's lateral GaN technology is enabling a new generation of efficient power conversion solutions across the 40V-650V voltage range. By delivering faster switching speeds, lower power losses, and higher power density than traditional silicon-based solutions, lateral GaN supports smaller, more efficient system designs with improved thermal performance. These advantages make it well suited for power supplies, industrial automation, data centers, telecommunications, consumer electronics, and electrification applications. As demand grows for higher efficiency and compact power architectures, lateral GaN provides a compelling technology platform for next-generation energy conversion and power management.

Qualcomm Presents:

Title: Qualcomm: Moving AI to the Edge and being more developer friendly

Abstract: Qualcomm is advancing edge AI by enabling intelligent processing directly on devices, addressing critical requirements around latency, privacy, power efficiency, and overall system cost. This approach leverages heterogeneous compute architectures—spanning CPU, GPU, and NPU—to match the right workload to the right engine while supporting real-world use cases such as vision, robotics, industrial automation, and IoT. At the same time, Qualcomm is prioritizing a more developer-friendly experience by reducing friction across the AI lifecycle, from model selection and optimization to deployment and fleet management. By closing the gap between proof-of-concept demos and scalable production systems, Qualcomm's edge AI ecosystem allows developers and partners to move faster, deploy securely, and bring AI-enabled products to market with greater confidence.

Silicon Labs Presents:

Title: Enable Accurate Distance Estimation Using Channel Sounding

Abstract: Explore the latest advancements in Bluetooth 6.0, with a special focus on Channel Sounding. Channel Sounding is a new standardized, interoperable, and secure method for distance measurement that is set to revolutionize a variety of applications. This session will also feature a comparative analysis of Channel Sounding with other ranging methods.

STMicroelectronics Presents:**Title: Migrating from Linux: ST Enables Efficient GUI Solutions**

Abstract: In this hands-on workshop, you'll discover how ST microcontrollers and TouchGFX work together to create rich, responsive graphical user experiences using remarkably little memory. By eliminating the need for external DRAM and avoiding the overhead of Linux-based systems, you can build simpler, more efficient products.

You'll also explore partial refresh technology—a breakthrough approach that dramatically reduces memory requirements while making system design even more elegant.

We'll provide everything you need to get started: ST NUCLEO development kits, preconfigured laptops, and a guided experience that takes you from concept to a working graphical interface.

Molex Presents:**Title: Innovation Driven by Automotive, Designed for Rugged Applications**

Abstract: "The challenges facing today's automotive industry—higher power, faster data, smaller packaging, and uncompromising reliability—are accelerating innovations that extend well beyond passenger vehicles. As electrification, software-defined architectures, automotive Ethernet, and advanced sensing become mainstream, connector technologies are evolving to deliver greater performance in increasingly demanding environments.

The seminar will showcase a broad range of Molex automotive technologies that are increasingly finding applications beyond passenger vehicles. Whether you're designing commercial vehicles, industrial equipment, agricultural machinery, robotics, marine systems, or other rugged applications, you'll discover how technologies developed for the automotive industry's demanding requirements can help improve reliability, performance, and enable scalable architectures to prepare your products for the next generation of connectivity."

9:30am to 10:20am – Session #2

Altera Presents:**Title: Robotics, Nvidia Attach, Power and Performance advantages and design productivity**

Abstract: Altera 3.0 represents the next iteration of the company making it the only pure-play FPGA company in the industry today. Learn about our value proposition in Robotics, Holoscan capabilities, and our investments into our architecture and tools you need to accelerate your design flow and beat your competition to market. Our Hyperflex and Tensor Flow DSP architecture allows for higher performance, lower power, and AI at the edge. This enables you to future proof your designs with edge AI and increase your productivity with higher utilization and ease of timing closure.

Harting Technology Presents:**Title: How to Select the Right Ethernet Connectivity Technology for Industrial Applications**

Abstract: Current landscape of industrial Ethernet

Overview of Ethernet Connectivity technologies

Comparing copper with Fiber Optic

Comparing 4&8 wire copper technology with Single Pair Ethernet

Pre-made cable assemblies vs assemblies made on-site

How to decide between both options

Ethernet communication in industrial applications

How to select appropriate connectivity technology for industrial devices and machinery based on given requirements and pain points

Identifying future applications for Single Pair Ethernet

Fiber Optic Ethernet in industrial markets

How to select appropriate Fiber Optic connectivity technology based on application and environment

Infineon Presents:**Title: Machine Learning (ML) for Engineers “What is ML at the EDGE”**

Abstract: Machine learning enables systems to learn from data and supports better decision-making, but successful outcomes depend on a structured process. This presentation introduces the core EDGE ML; Data Collection, Preprocessing, Model Training, Model Evolution, and Model Deploy. Attendees will gain a practical understanding of how these processes work together to build reliable machine learning solutions. Through real-world examples and best practices, the session provides a clear foundation for understanding the machine learning workflow and applying its principles to solve business and organizational challenges.

Littelfuse Presents:**Title: TVS Technology Trends: Higher Power, Lower Clamping, Better Protection**

Abstract: As electronic systems continue to evolve toward higher power density, faster switching speeds, greater connectivity, and increased integration, the challenges associated with transient voltage threats are becoming more severe. Traditional surge protection approaches are being pushed to their limits as designers balance reliability, performance, efficiency, size, and cost. This seminar explores the changing transient protection landscape, covering common electrical threats such as ESD, EFT, and surge events, and compares the strengths and limitations of TVS diodes to MOVs, GDTs, and SIDACTor® technologies. Attendees will gain practical insight into TVS diode operation, selection criteria, and design considerations, while examining how market trends are driving the development of innovative high-power and low-clamping TVS technologies that provide enhanced protection for today's increasingly sensitive electronic systems.

Rohm Semiconductor Presents:**Title: ROHM's 5th Generation SiC MOSFETs for highly efficient automotive and industrial power systems**

Abstract: ROHM has developed its 5th Generation EcoSiC™ power MOSFETs, designed to meet the demanding requirements of electric vehicles, solar inverters, energy storage systems, AI data centers, and other industrial power applications. This latest generation of SiC technology delivers industry leading efficiency, achieving up to a 30% reduction in on resistance at elevated temperatures compared to existing products. These improvements enable component and system level downsizing without compromising output power capability.

ROHM's expanded portfolio of EcoSiC™ discrete and module packages supports a wide range of design requirements, including system compactness, component compatibility, thermal performance, cost competitiveness, and ease of assembly.

Join this seminar to explore the latest advancements in SiC power devices and discover how they enable higher performance, more efficient power electronics solutions.

u-blox Presents:**Title: Rapid and Secure Bluetooth and Wi-Fi Connectivity for IoT Devices**

Abstract: "Bringing wireless connectivity to embedded products can be one of the most challenging aspects of IoT development, often requiring expertise in networking, security, cloud protocols, and wireless communications. This session explores how u-connectXpress enables developers to rapidly add Bluetooth and Wi-Fi connectivity through a simple AT command interface, eliminating much of the complexity associated with traditional wireless stack integration.

Attendees will learn how to reduce development time, simplify security implementation, and accelerate product deployment using built-in support for secure cloud communications and industry-standard wireless technologies. The presentation will also cover strategies for improving design portability across hardware platforms, reducing maintenance effort, and enabling scalable IoT solutions for industrial, medical, smart building, and connected-device applications.

Target Audience: Embedded engineers, IoT developers, system architects, and product managers seeking faster, more secure wireless product development."

10:30am to 11:20am – Session #3

Critical Link Presents:**Title: Critical Link SOMs: Implementing AI at the Edge, Embedded Linux Security, & AMP**

Abstract: Building on last year's introduction to AI at the edge and AMP, this seminar takes a deeper look at real-world implementation. We will cover practical strategies for getting started with edge AI using Altera and Qualcomm-based SOMs, including a live Agilex 5 demo, techniques for interprocessor communication and resource partitioning across multi-processor and FPGA systems, and an attack surface analysis of embedded Linux systems.

Microchip Presents:**Title: Do More with Less: The Next Generation of Cost-Optimized MCUs**

Abstract: Discover how configurable logic, integrated security, low-power architectures, and AI-assisted development tools are helping engineers build more capable embedded systems faster and at lower cost.

Lumileds Presents (added 2026-09-30):

Title: Lighting the Future of Transportation

Abstract: This course introduces the breadth of Lumileds LED technologies shaping today's transportation lighting landscape. Attendees will explore how Lumileds delivers complete exterior lighting solutions for cars, trucks, two wheelers, heavy duty and specialized vehicles, leveraging high performance LUXEON emitters, modules, and integrated systems. We will examine standards driven requirements (SAE, AEC, NHTSA), the evolution of LED firsts in automotive lighting, and the expanding role of LEDs in styling, safety, signaling, and emerging mobility applications. Through real market examples and technology roadmaps, participants will gain a clear understanding of how Lumileds innovations—known for robustness, efficiency, and OEM grade quality—enable differentiated designs, faster integration, and superior performance across transportation platforms.

Nexperia Presents:

Title: Application Specific Power MOSFETs (ASFETs) and Their Unique Performance Advantages

Abstract: Power MOSFETs are not all equivalent. Various device technologies provide unique performance advantages. Nexperia has developed Application Specific MOSFETs that are a better solution for common applications. These include Avalanche performance, Hot Sway, Motor Control, Transient Linear Mode and others. This seminar provides an overview of these ASFET families and their unique qualities.

Vishay Intertechnology Presents:

Title: Optimizing Industrial Motor Drive and Motor Control System Performance with Vishay Components: Enabling Higher Efficiency, Reliability, and Power Density

Abstract: Industrial motor drives are essential to automation, enabling precise speed control, energy efficiency, and reliable operation in robotics, pumps, HVAC, compressors, and material handling.

This session will cover key component selection considerations for power conversion, inverter stages, current sensing, EMI mitigation, circuit protection, thermal management, and power quality. It will highlight Vishay solutions including SEMIs, Passives, and Isolation products.

Attendees will learn Vishay's latest technologies to improve efficiency, reliability, and power density in next-generation industrial motor control designs.

Phoenix Contact Presents:

Title: Overcoming connectivity challenges for robotic integration

Abstract: Robotics relies on robust connectivity for real-time control, data exchange, and system integration. This encompasses internal electrical and electronic connections as well as communication between robots and external devices. Performance is often constrained by latency, bandwidth limitations, and network instability, particularly in autonomous vehicles, industrial automation, and multi-robot environments. Advances in connector design and emerging technologies offer improvements, yet achieving reliable, low-latency connectivity in dynamic conditions remains a critical challenge for fully autonomous and collaborative robotics.

11:40am to 12:20pm – Arrow Sessions

eInfochips Presents:

Title: EdgeAI Systems – From POCs to Production

Abstract: "Edge AI is moving beyond POCs to production deployments, but building scalable, reliable, and secure Edge AI systems requires more than selecting the right AI model."

This session explores the key engineering challenges of building production-ready Edge devices, highlighting the hardware platforms and software frameworks required for production-scale deployments. eInfochips will share real-world engineering experience and demonstrate how organizations can leverage System-on-Modules (SOMs) and software accelerators to reduce development complexity, shorten time-to-market, and successfully deploy Edge AI solutions across manufacturing, healthcare, transportation, and industrial automation."

SiliconExpert Presents:

Title: How Unified Intelligence Turns Supply Chain Volatility into Competitive Advantage

Abstract: Geopolitical instability, AI-driven demand surges, lifecycle changes, supplier concentration, and market volatility have permanently raised the stakes for electronics manufacturers. The companies that win in this environment act faster and decide smarter, and that requires supply chain, compliance, and engineering to operate from a single source of truth.

Most organizations are not there yet. Fragmented data, low AI confidence, reactive sourcing, and late-arriving signals create bottlenecks that slow every team down and widen the window of exposure. This is an alignment problem as much as a technology problem.

When AI serves as the operating layer of the supply chain, with unified intelligence spanning supply chain, procurement, engineering, and compliance, plus continuous global-event monitoring and mitigation tracking, every team responds to the same prioritized signal at the same time. Immediate BOM-level visibility, AI-summarized risk context, validated alternates, and coordinated procurement response⁴ become the new standard.

This seminar walks through what that integrated model looks like in practice, including live scenario walkthroughs and a clear framework for building resilience that keeps pace with global dynamics.

12:30pm to 1:20pm – Session #4

Digi International Presents:

Title: How to Scale Edge AI from Development to Deployment in Industrial Inspection

Abstract: Industrial teams face mounting challenges moving AI models from development to real-world edge deployment, where hardware complexity, accelerator diversity, and traditional centralized inspection systems can slow time to market and limit scalability. This webinar uses a real-world optical inspection example from food production to demonstrate how edge AI and cloud-based services work together across the full lifecycle — from training and deployment to monitoring, retraining, and over-the-air updates — simplifying the path to scalable, continuously improving edge AI systems.

Honeywell Sensing Presents:

Title: Discover How Advanced Pressure & Flow Sensing Can Transform Your Medical ,Industrial and Aerospace Applications

Abstract: "Join us for an engaging session exploring how next generation pressure and liquid flow sensors are powering innovation across medical, industrial, and aerospace markets. These sensors play a vital role in the accuracy, safety, and reliability of critical systems—from life saving medical devices to high performance industrial and aerospace equipment.

You'll see how our sensing technologies enable smarter, more dependable operation in applications such as dialysis equipment, blood analysis systems, centrifuges, oxygen and nitrogen gas distribution, HVAC units, data storage, process controls, industrial machinery, pumps, and robotics"

Lattice Semiconductor Presents:

Title: Utilizing FPGA's as Companion chips in embedded applications

Abstract: As MCU/MPU demands grow in embedded and edge AI applications, designers can offload tasks to small, low-power, low-latency FPGAs. This seminar shows how adding FPGAs expands system headroom and enables new features without requiring major changes to existing processors from Qualcomm, NXP, ST, Nvidia, and others.

MCC Semiconductor Presents:

Title: MOSFET selection for BESS: Battery Energy Storage System

Abstract: MOSFET selection for BESS is a critical activity for back-to-back, matrix cell-selection, flyback converter, and balancing enabling circuit blocks. As global demand for energy increases also the need for efficient and reliable BESS. These systems will require to have balancing strategies that can be passive (single cell energy dissipation) or active (cell-to-cell energy transferring).

Melexis Presents:**Title: Ultracompact 12 / 24 / 48 V Controls**

Abstract: Learn how Melexis sensors and application-specific microcontrollers can help solve some of the most common mechatronic hardware challenges: space constraints, quiescent power, robust communication, and cost. Melexis products bring 37 years of automotive expertise and production-proven 48 V success to your design. Whether you're producing a generator, pump, valve, fan, handheld tool, or robotic actuator, Melexis microcontrollers offer the smarts you need to offer true system solutions in a tiny package. Not looking to write firmware? Learn about our pragmatic and flexible "code-free" approach.

Taiyo Yuden Presents:**Title: Conductive Polymer Hybrid Aluminum Electrolytic Capacitors (Polymer Hybrid Capacitor)**

Abstract: "TAIYO YUDEN conductive polymer hybrid aluminum electrolytic capacitors combine the benefits of liquid electrolytes and solid conductive polymers. They offer ultra-low Equivalent Series Resistance (ESR), high ripple current capabilities, and a long operational life at high temp up to 150 °C 2000 hours.

Taiyo Yuden will introduce Polymer Hybrid Capacitor and highlight what the trends are in making quality Hybrid Capacitor. We will contrast our different product families so that you can select the best performance device to support your application and we will cover our website tool for component selection called COMPAS."

1:30pm to 2:20pm – Session #5

Adam Tech Presents:**Title: Connecting the Future: Transportation & Energy Management**

Abstract: Topics will cover automotive, transportation, and transportation adjacent applications, in addition to industrial topics including energy management with complimentary building control / security applications reviewed.

Advanced Energy Presents:**Title: Power Supply System EMI Fundamentals**

Abstract: As power electronic systems continue to operate at higher switching frequencies and power densities, electromagnetic interference (EMI) has become a critical design consideration affecting product performance, regulatory compliance, reliability, and time-to-market. This training session provides a practical introduction to EMI concepts with a focus on power supply applications and real-world mitigation techniques.

The session begins with an overview of EMI fundamentals, explaining what EMI is, how it is generated, and why reducing EMI is essential for modern electronic systems. Participants will gain an understanding of the electric and magnetic field mechanisms that create electromagnetic noise and how these fields couple into surrounding systems. Building on these fundamentals, the course examines EMI behavior within power supply systems, covering both conducted and radiated emissions. Special attention is given to the the dominant noise paths found in switching power supplies, and the mechanisms by which these disturbances propagate through equipment and system interconnections.

The training also reviews key EMI compliance standards and regulatory requirements commonly encountered in industrial, medical, and commercial applications. Participants will learn how emissions are measured, interpreted, and evaluated against applicable standards.

The session will include an overview of Advanced Energy's EMI resources, facilities, and support capabilities, including laboratory infrastructure, compliance engineering expertise, and field and customer support.

Allegro MicroSystems Presents:**Title: Gate Driver Architectures in Modern Motion Control and Power Conversion**

Abstract: As motor control and power conversion applications push the boundaries of power density, efficiency, and switching frequency, selecting the optimal gate driver is critical. This technical seminar provides a guide to selecting and designing gate driver stages across diverse voltage and power boundaries. We will explore the critical trade-offs between Silicon MOSFETs, Silicon Carbide (SiC), and Gallium Nitride (GaN) power switches.

Participants will gain a deep understanding of driver topologies, comparing low-voltage gate drivers (<100 V) used in compact robotic joints with high-voltage isolated gate drivers (>100 V) engineered for industrial power supplies and isolated power systems. Key system integration challenges will be analyzed, such as Common Mode Transient

Immunity (CMTI), Elimination of External Isolated Power Supplies, mitigating extreme voltage transients and EMI, and reducing system footprint through high density power and sensing integration.

Diodes Inc. Presents:

Title: Solve common design problems with programmable, mixed-signal uASICs from Diodes Inc.

Abstract: Learn how to save board space, power, and costs with Diodes' configurable uASICs solutions for industrial and consumer applications. In this session we will address common design needs like power sequencing, watchdog supervisors, waveform generators, voltage monitors, OVP/UVP and OCP functions, and the implementation of simple and complex logic functions in a programmable, mixed-signal ASIC. We will show design tool examples and demonstrate how simple it is to implement a solution using uASICs from Diodes Inc.

Nuvoton Presents:

Title: "Smart Buzzer® – Transforming Piezo Buzzers into Meaningful Audio Alerts"

Abstract: "Stop settling for beeps that users ignore—discover how Nuvoton's Smart Buzzer® turns conventional piezo buzzers into intelligent voice-capable notification systems that improve safety, usability, and product differentiation while preserving the low cost, power efficiency, and simplicity industrial designers demand."

Vantron Technology Presents:

Title: From Edge to Cloud: Edge AI & Networking Solution for Remote Industrial Sites

Abstract: Explore Vantron's edge-to-cloud solution combining industrial control, embedded computing, IoT networking and cloud management. Through a remote energy deployment, this session showcases how AI-enabled Wi-Fi HaLow IP cameras, HaLow routers, and NXP i.MX94 gateways work together to deliver smart edge processing, kilometer-scale connectivity, secure IIoT monitoring, and seamless cloud integration.

* Subject to change without notice